Casey Barton

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PROFILE

I build the machine that decides whether an agent is allowed to ship: evaluation harnesses, trace intelligence, and benchmark suites that turn agent behavior into measurable, improvable facts. Every agent I ship runs under an eval discipline first — golden sets, behavioral assertions, replay loops, and regression gates before production. Shipped agents and agentic applications to production, then tuned them for accuracy against real datasets.

Three years as a founder running an evidence-grade agent control plane from Honolulu: hand-rolled multi-agent loops with function calling, structured outputs, typed handoff contracts, durable recoverable runs, **65** MCP connectors, and full run logging so any execution can be reconstructed. Remote-first; experimentation-to-production fluency.

CORE COMPETENCIES

Engineering: TypeScript · Python · Node.js · SQL · Bash · REST APIs · containers · GitHub Actions (CI/CD) · Supabase · Postgres

AI & Evaluation: LLM evaluation harnesses · trace intelligence · benchmarking & experiments · autoevals · memory systems · RAG pipelines · function calling · structured outputs · hand-rolled multi-agent loops · guardrails · fail-closed recovery · MCP protocol · prompt & context engineering

Production: production framework engineering · observability & run logging · latency & token-cost benchmarks · CI regression gates · decision audit receipts · experimentation discipline · documentation

EXPERIENCE

Founder / Applied AI Systems Architect 2023 – Present

GlacierEQ — Honolulu, HI

- Ship agents and agentic applications to production, then tune them for accuracy: prompt and configuration changes validated by eval harnesses with replay loops before anything touches production — an agent that cannot be measured cannot be trusted.
- Build trace intelligence into every system: full logging of every prompt, tool call, and decision, so any run can be reconstructed and debugged from its trace — failed evals are diagnosed, not argued about.
- Benchmark agent primitives against datasets and iterate on the findings: run experiments on memory, retrieval, and orchestration behavior, investigate where they perform well and where they fall short, and turn results into configuration improvements.
- Run an evidence-grade agent control plane across a **1,395-repository** estate: typed state transitions, human approval gates on consequential actions, durable state that survives a failed run, recoverable runs — backed by a **989-test** suite, **62/62** verification checks green.
- Integrate **65** MCP connectors — authoring MCP integrations that expose tools and structured resources to any MCP-compatible system — plus APIs, webhooks, and data pipelines, with evidence boundaries and audit receipts.

- Design evaluations, not spot checks: golden suites with behavioral assertions, judge criteria with written rubrics, and CI gates that block a regression from merging. Fail-closed by default, never guess.

EDUCATION

University of Hawaii at Manoa 2016
Bachelor of Science

AWS Cloud Institute 2025
Cloud Computing program

Colorado State University
Undergraduate coursework

SELECTED PROOF

Job-App Helix — evaluation-gated campaign engine

- Attributable campaign engine with immutable evidence models, stable finding codes, and fail-closed **GO / NO-GO** decisions; recovery restricted to contingency evidence declared before reassessment.
- Verification gates on every agent action: no agent completion claim without independent provider readback — the eval gate decides, not the agent.

GlacierEQ estate — production agent systems

- Durable patterns for claim tracking, evidence boundaries, audit receipts, and human-in-the-loop control across the full estate.
- Per-task latency instrumentation and decision audit logs across agent runs — the tracing practice that makes benchmarking honest.

SKILLS

Languages: TypeScript · Python · SQL · Bash · JavaScript / Node.js

AI / Evaluation: LLM evaluation harnesses · trace intelligence · benchmarking & experiments · autoevals · memory systems · guardrails · MCP protocol · typed handoff contracts · prompt & context engineering

Production: experimentation discipline · observability & run logging · CI regression gates · fail-closed recovery · documentation & handover

Evidence boundary: portfolio repositories are independent work. Domain-inspired demonstrations are not claims of company affiliation, production deployment, or certification.